CASUS
CENTER FOR ADVANCED SYSTEMS UNDERSTANDING

ANIMOVE 2025

Movement speed estimation

Using the ‘ctmm’ R package

Inês Silva

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Uniwersytet Wrocławski

MAX PLANCK INSTITUTE OF MOLECULAR CELL BIOLOGY AND GENETICS

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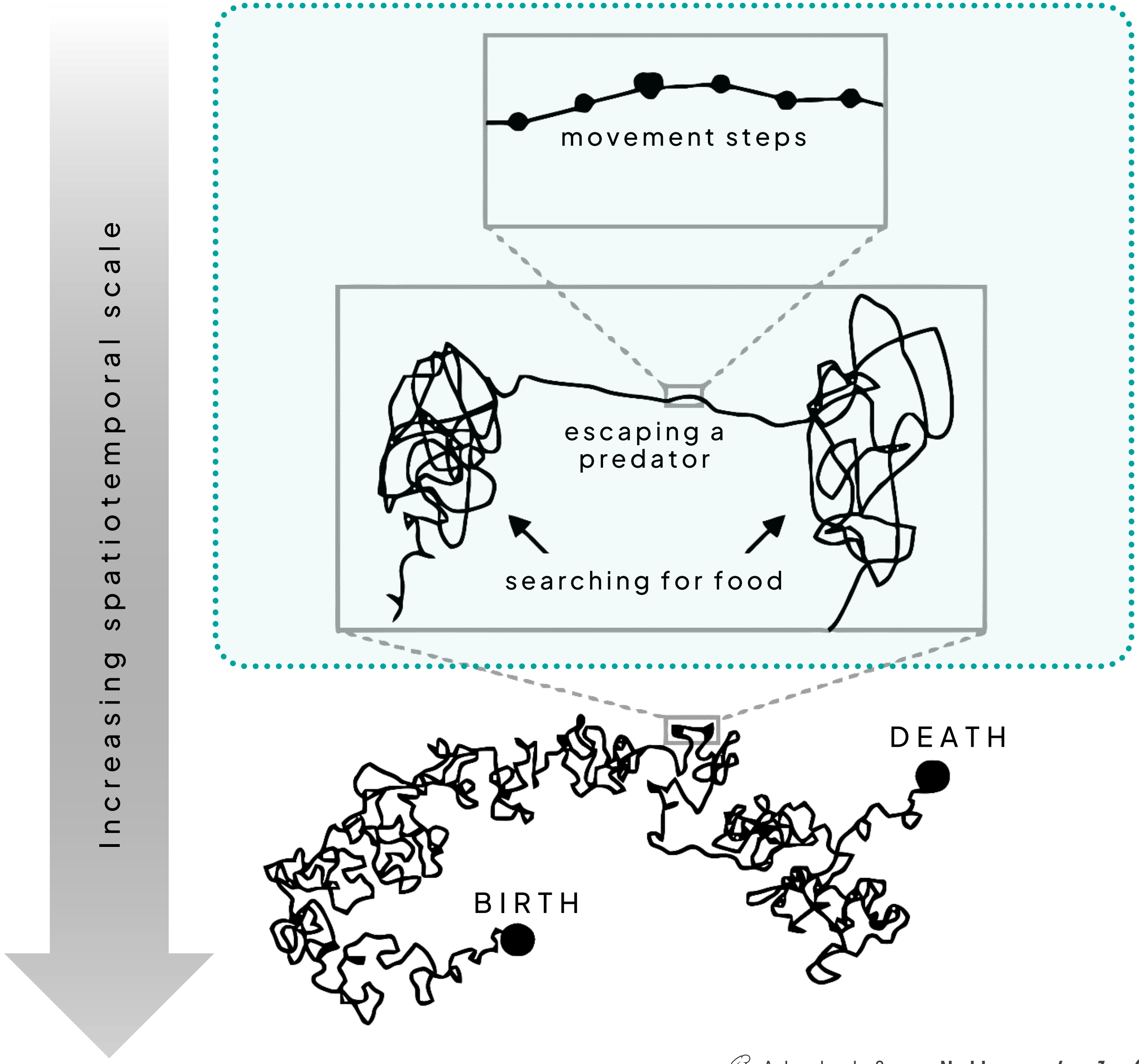
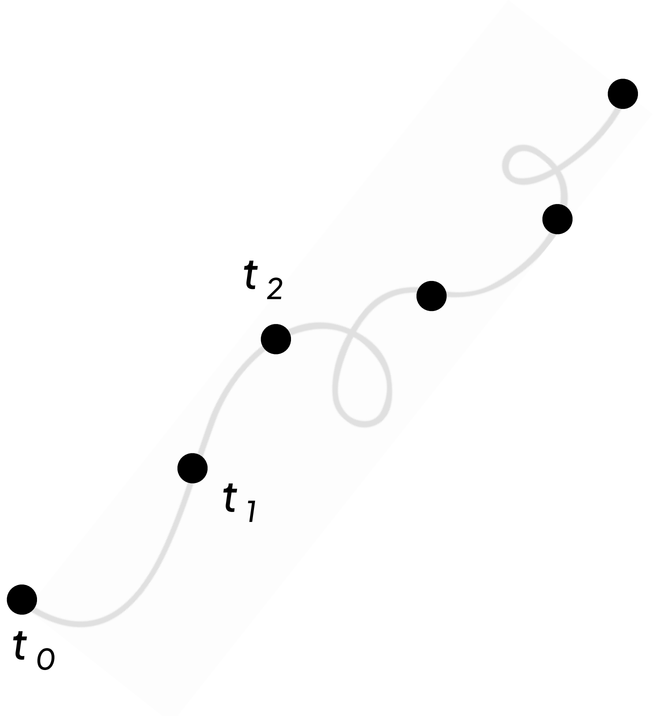
SACHSEN

Diese Maßnahme wird mitfinanziert mit Steuermitteln auf Grundlage des vom Sächsischen Landtag beschlossenen Haushaltes

Speed- and distance- related metrics provide quantifiable links between ***behavior* and *energetics***, can inform on risk/reward tradeoffs or as signals of ***anthropogenic disturbance***.



Adapted from Nathan *et al.* (2008)

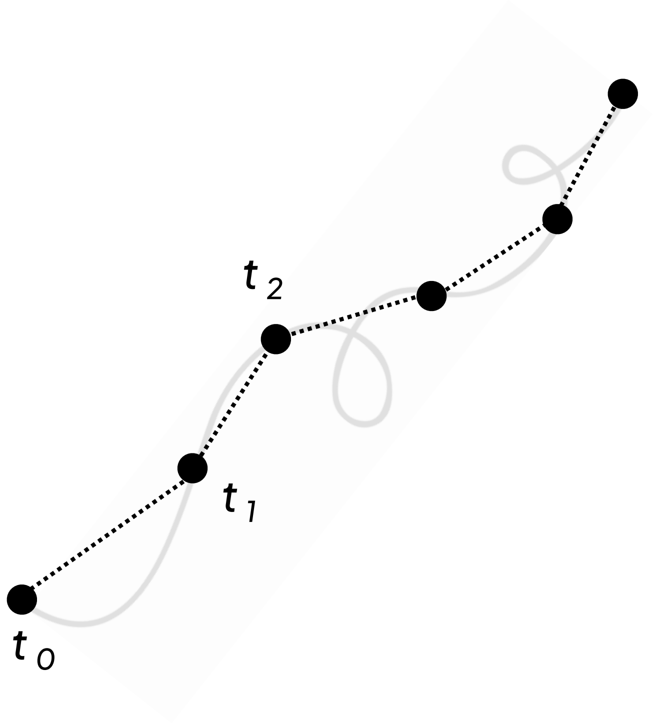


Speed and distance traveled are among the most routinely estimated metrics from animal tracking data.

Usually estimated by summing the ***straight-line distance (SLD)*** between location estimates…

$$\hat{d} = |\Delta \mathrm{r}| = \sqrt{\Delta x^2 + \Delta y^2}$$

... and divide that by Δt if speed if the desired metric.



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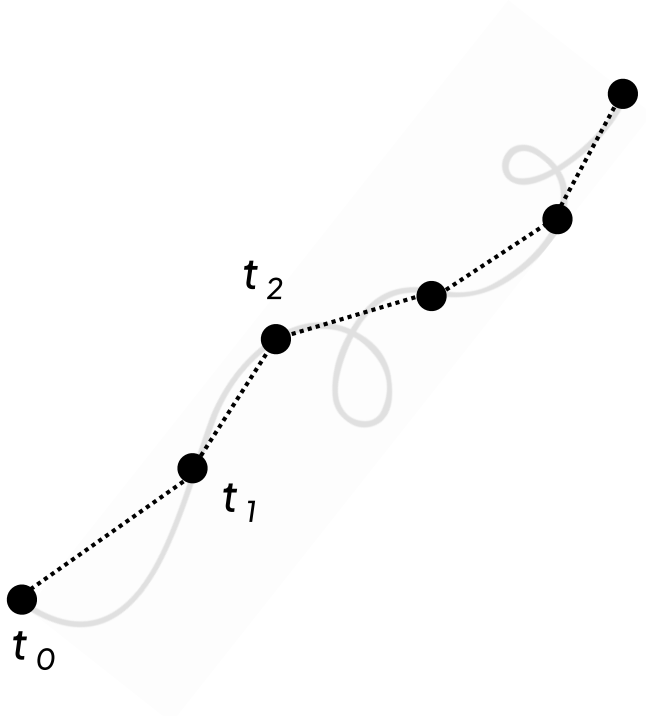

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Although SLD is easy to quantify,
it is also ***heavily biased***.

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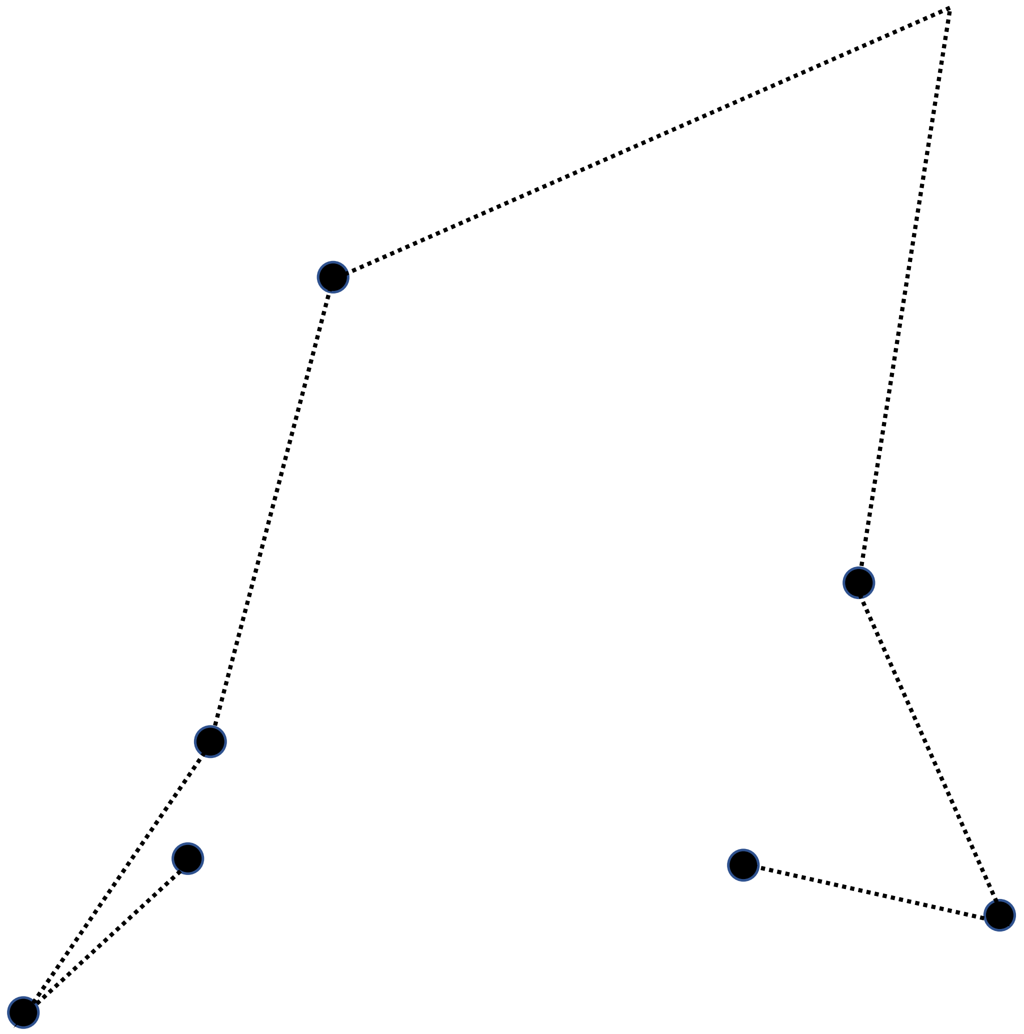
Estimating speed and distance



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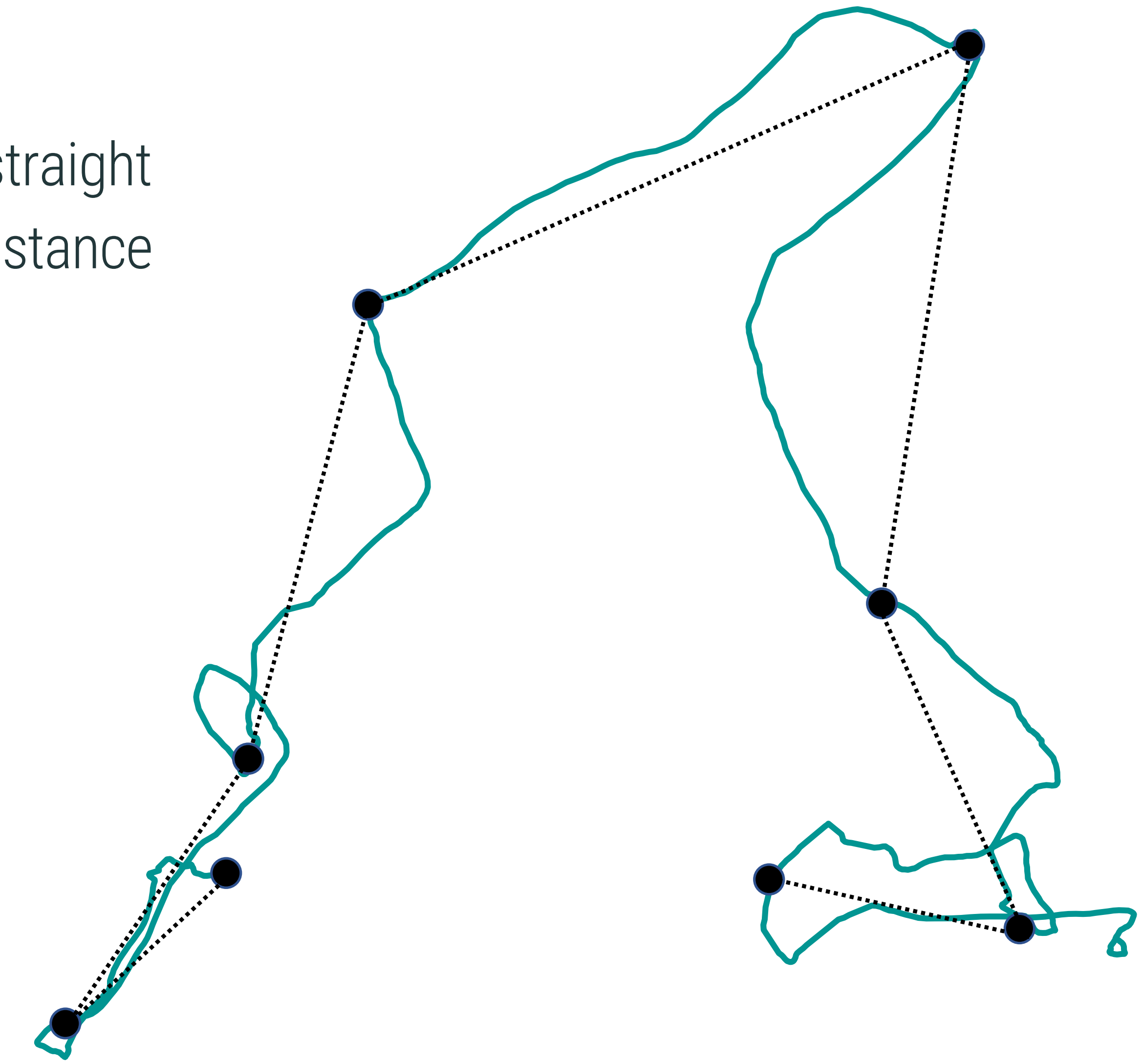
Unless the animal moved in a perfectly straight line, this will always ***underestimate*** distance traveled.

People have known this for decades:

Bovet & Benhamou, 1988;
Reynolds & Laundre, 1990;
Rooney *et al.*, 1998;
Turchin, 1998;
De Solla *et al.*, 1999;
Rowcli e *et al.*, 2012;
Sennhenn-Reulen *et al.*, 2017



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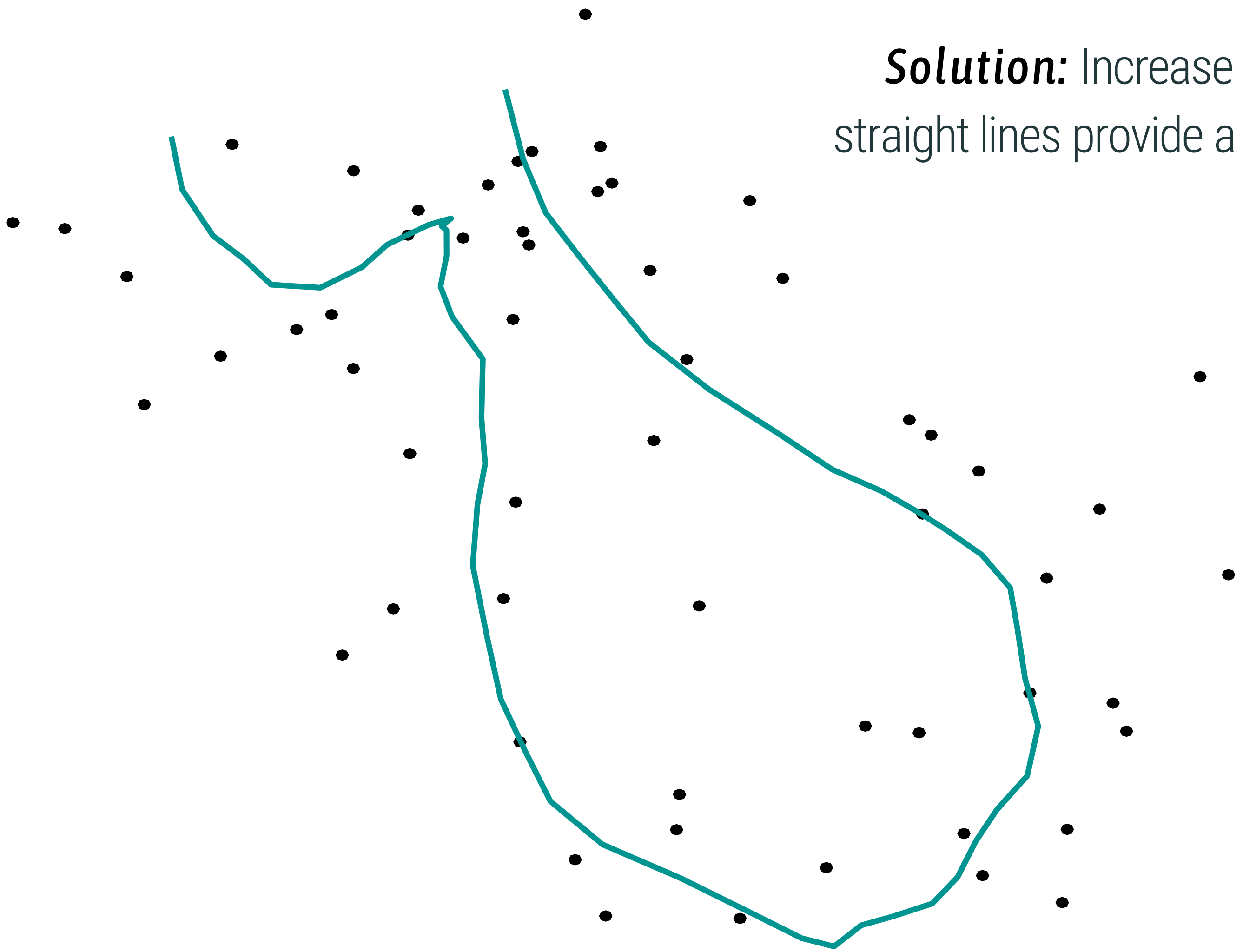
Solution: Increase the sampling frequency until the straight lines provide a better ***approximation*** of a curve?

Riemann sum



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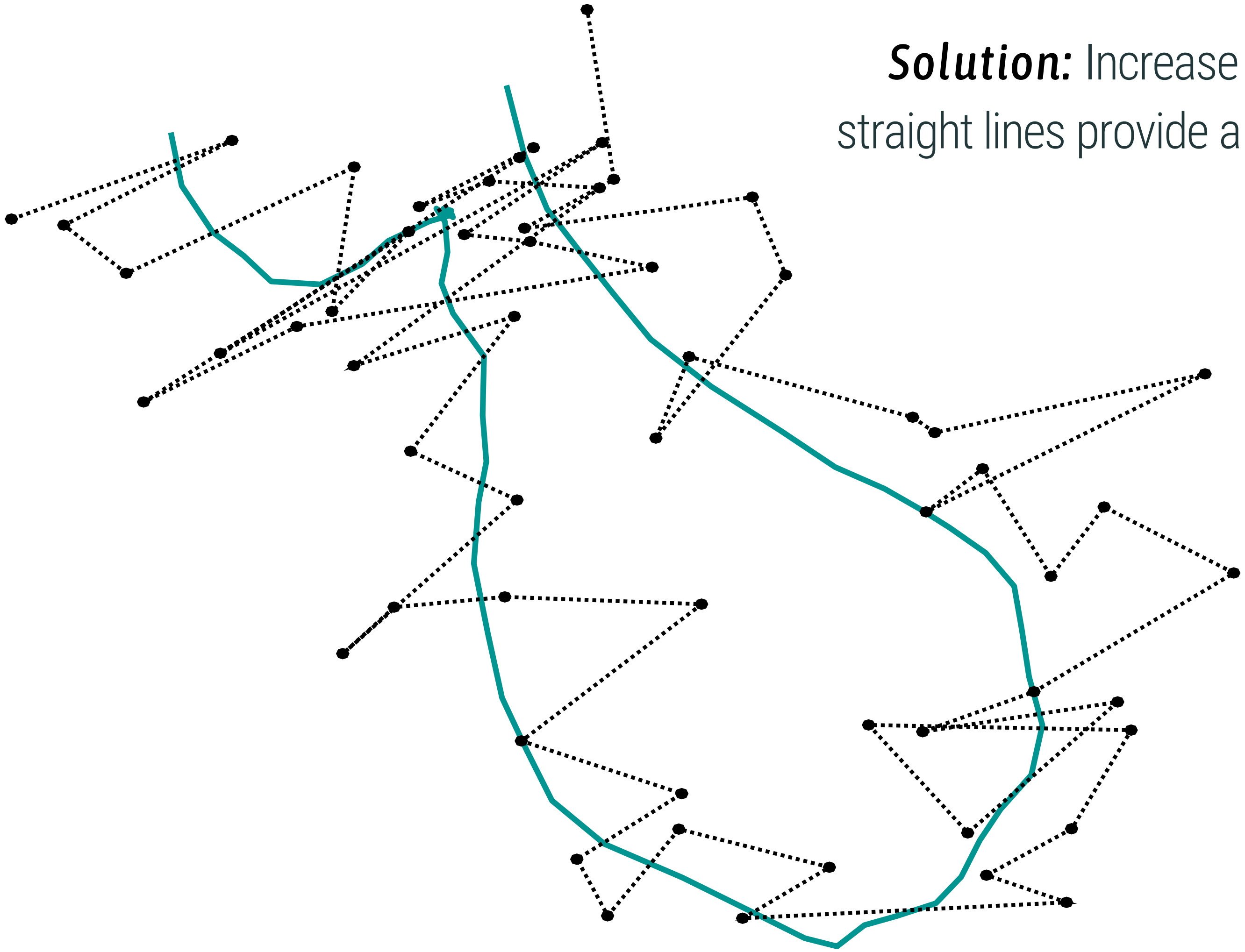

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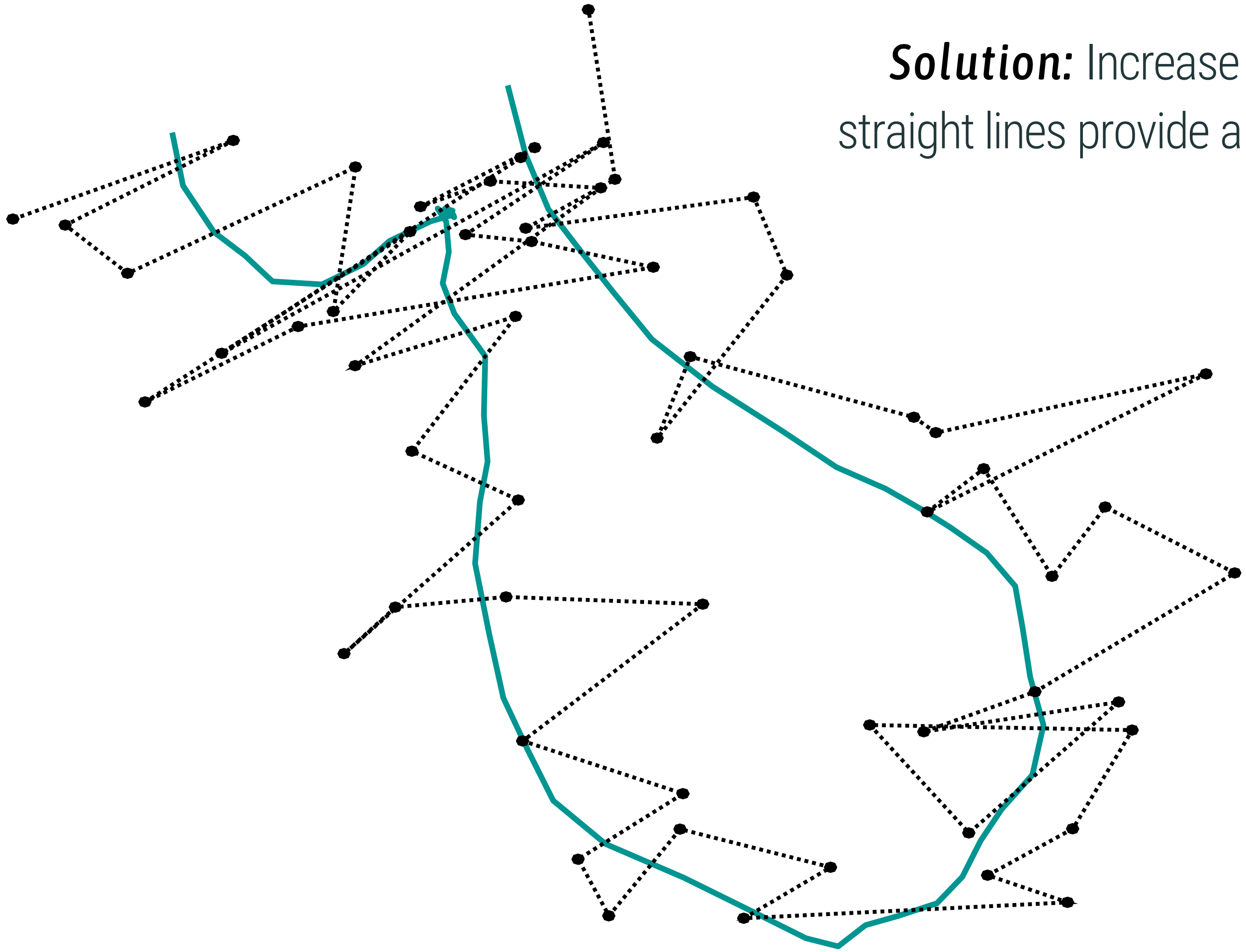

Solution: Increase the sampling frequency until the straight lines provide a better ***approximation*** of a curve?

Riemann sum

However,

If error is uncorrelated in time, estimates converge to ***infinity*** with infinite sampling frequency $(\Delta t \rightarrow 0)$.

Why? The actual distance traveled by the animal goes to ***0*** in the limit where $\Delta t \rightarrow 0$, but the magnitude of uncorrelated measurement error is ***independent of t***.

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Solution: Increase the sampling frequency until the straight lines provide a better ***approximation*** of a curve?

Riemann sum

$$\lim_{\Delta t \to 0} \left| \frac{\Delta}{\Delta t} \underbrace{(\mathbf{r} + \mathbf{error})}_{\text{observable}} \right| = \infty$$

Few people have been thinking about this
e.g., Ranacher *et al.*, 2015

However,
If error is uncorrelated in time, estimates converge to ***infinity*** with infinite sampling frequency $(\Delta t \to 0)$.

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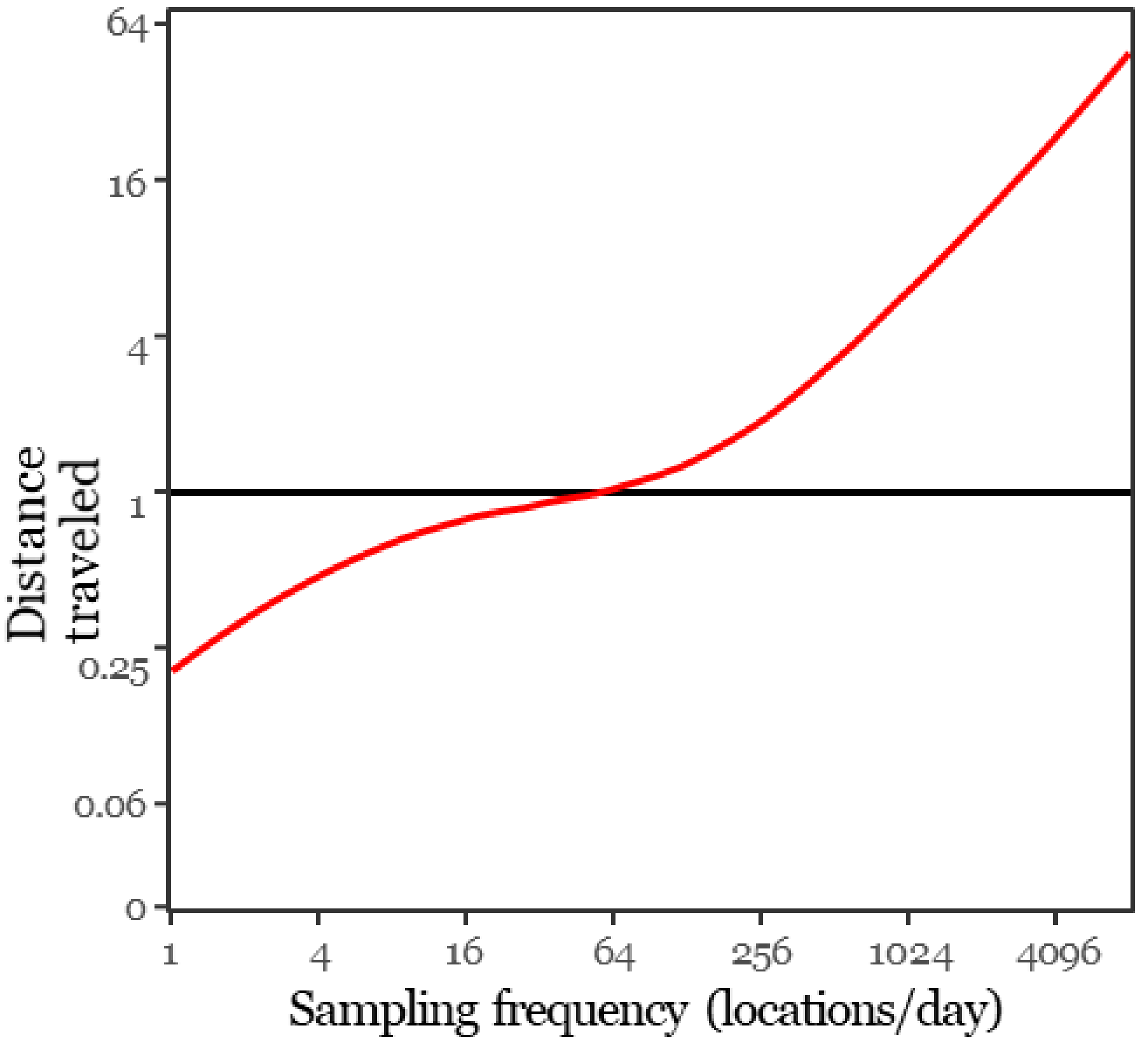
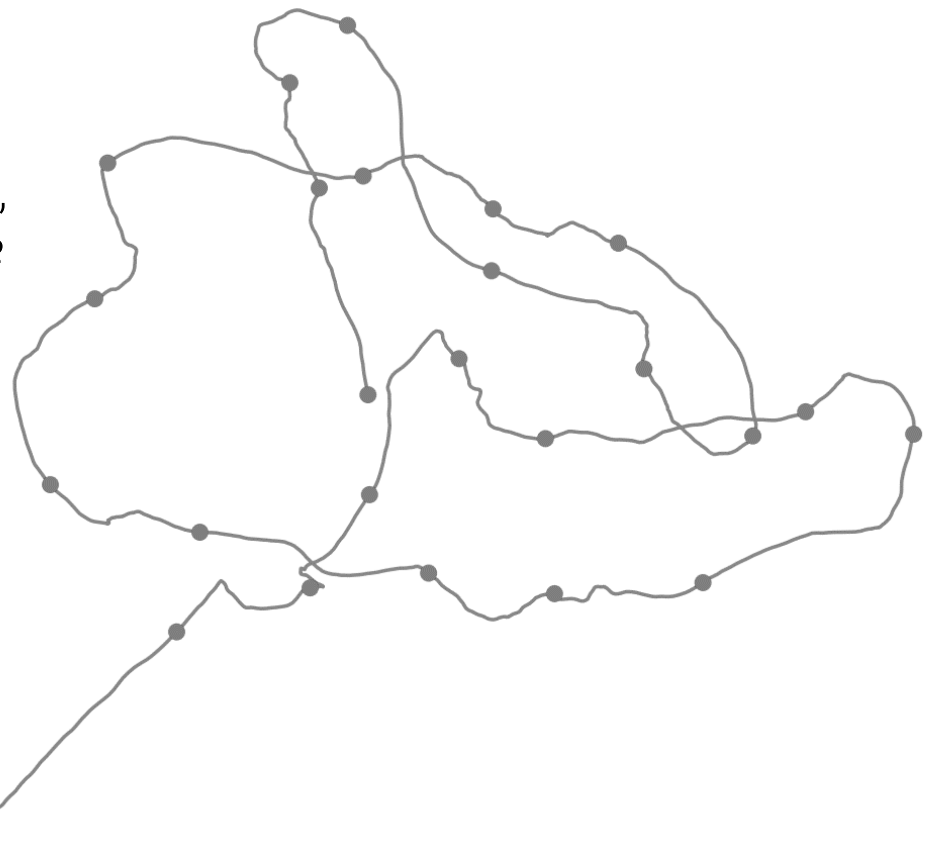


Continuous-time speed and distance (CTSD)

Noonan *et al.* (2020)

Select best-fit movement model, so we can repeat over ***multiple rounds of simulations***.



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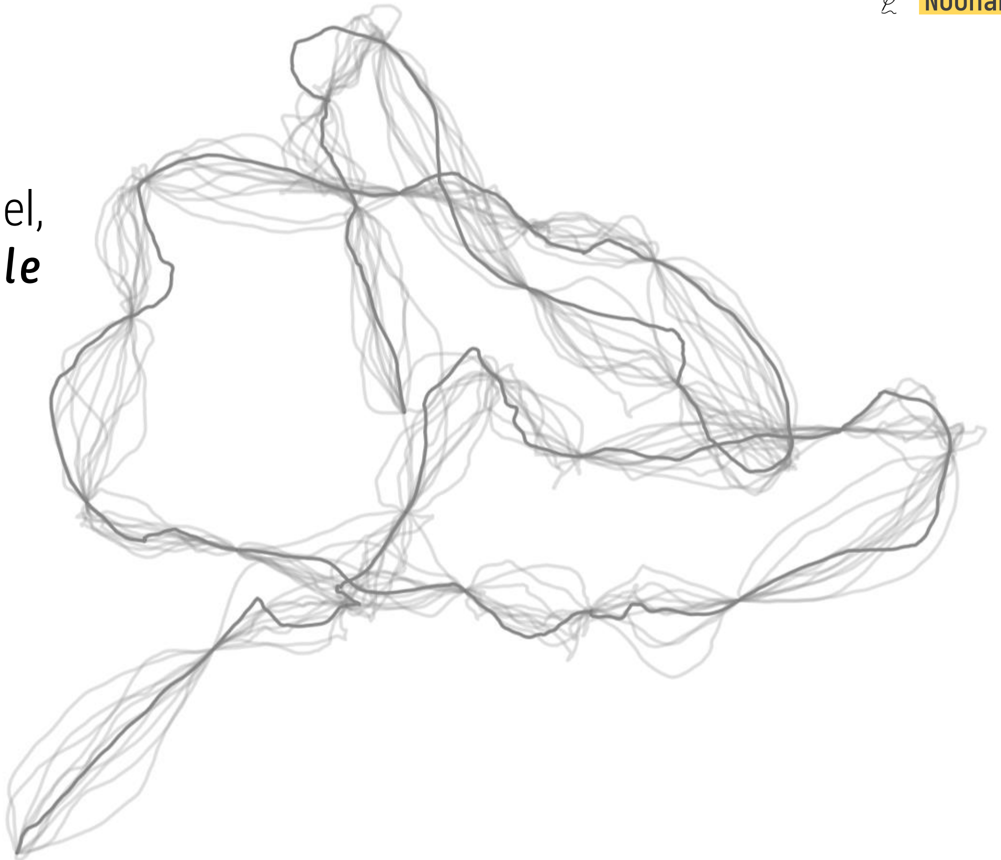


Continuous-time speed and distance (CTSD)

Noonan *et al.* (2020)

Select best-fit movement model, so we can repeat over ***multiple rounds of simulations***…

…Until we get a ***point estimate***, and variance around the estimate (for the ***confidence intervals***).

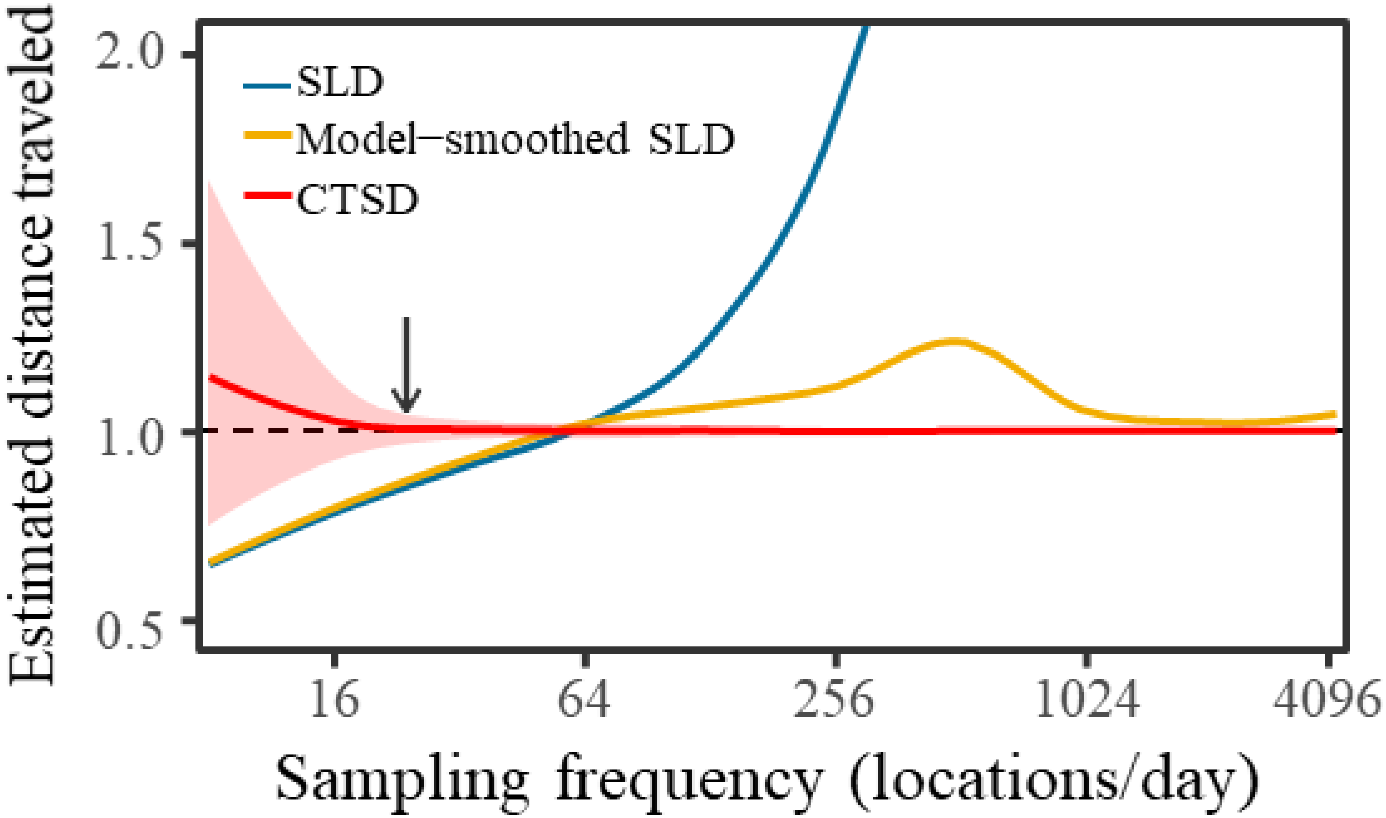
Continuous-time speed and distance (CTSD)

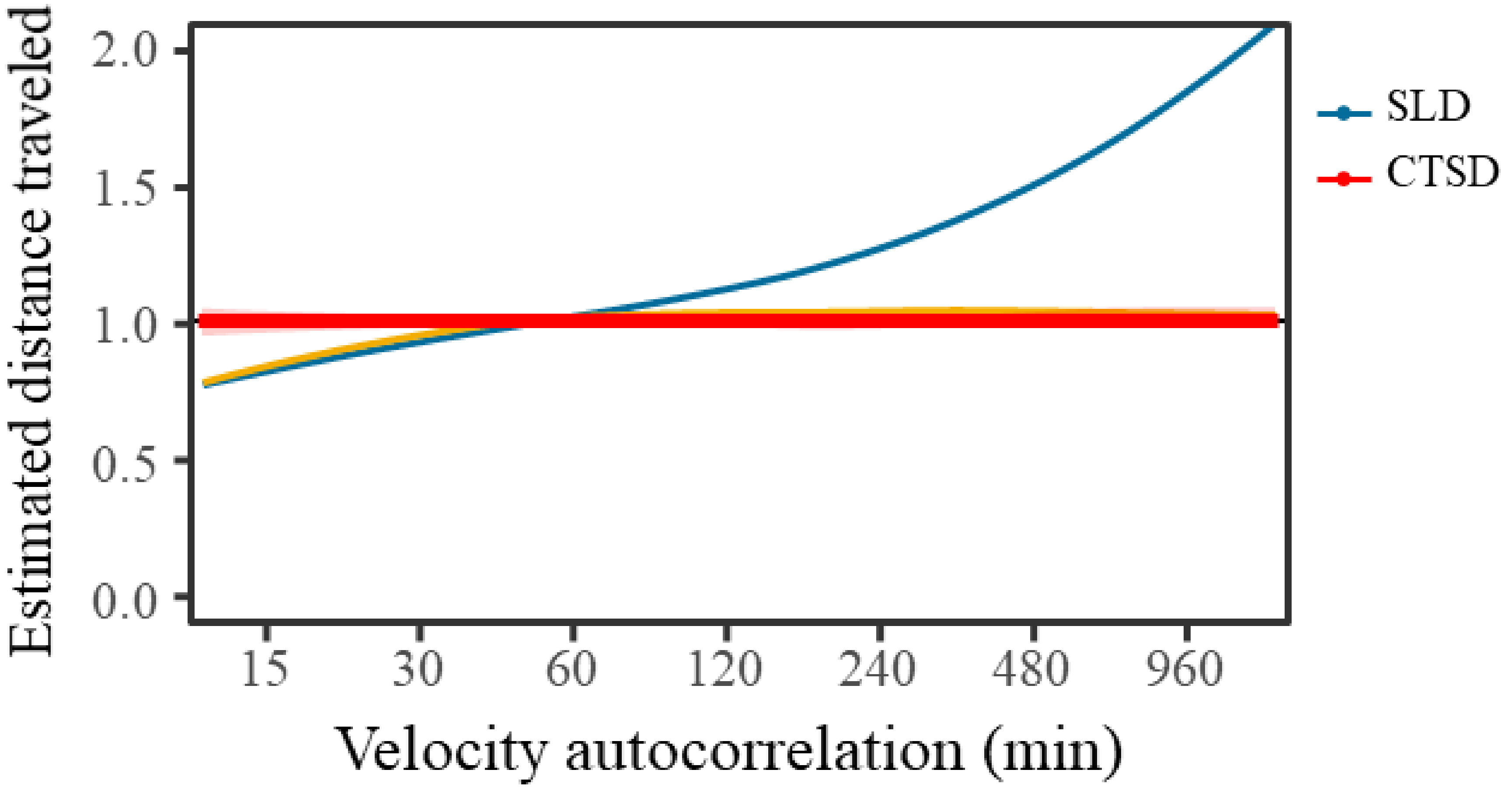
Noonan *et al.* (2020)

	Autocorrelation			
Model	Position	Velocity	Restricted	Parameters:
IID	No	No	Yes	τ = NULL
BM	Yes	No	No	$\tau = \infty$
OU	Yes	No	Yes	$\tau = \tau_p$
IOU	Yes	Yes	No	$\tau = \{\infty, \tau_v\}$
OUF	Yes	Yes	Yes	$\tau = \{\tau_p, \tau_v\}$

Requires a ***correlated velocity model.***

Estimating speed and distance





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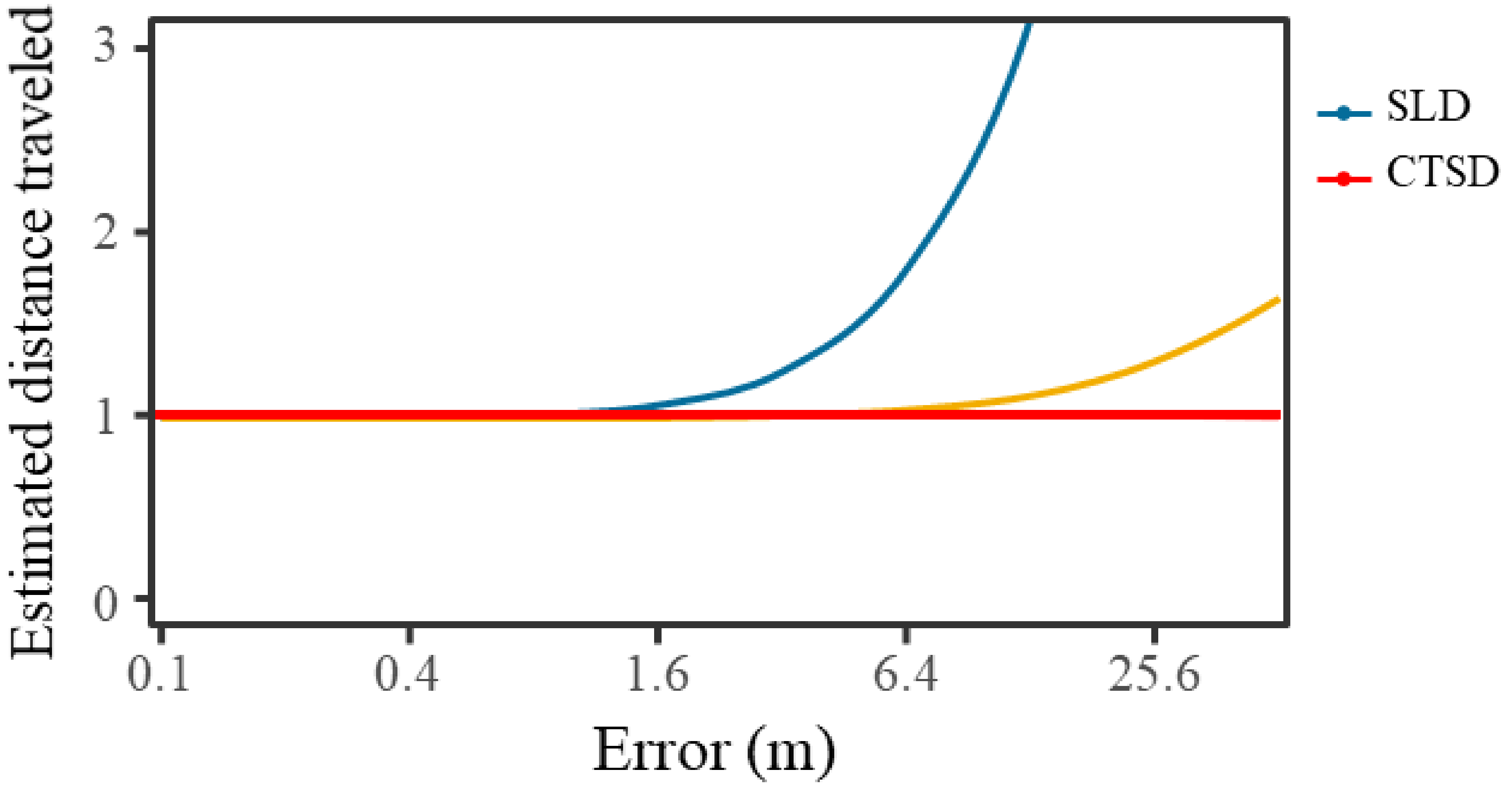
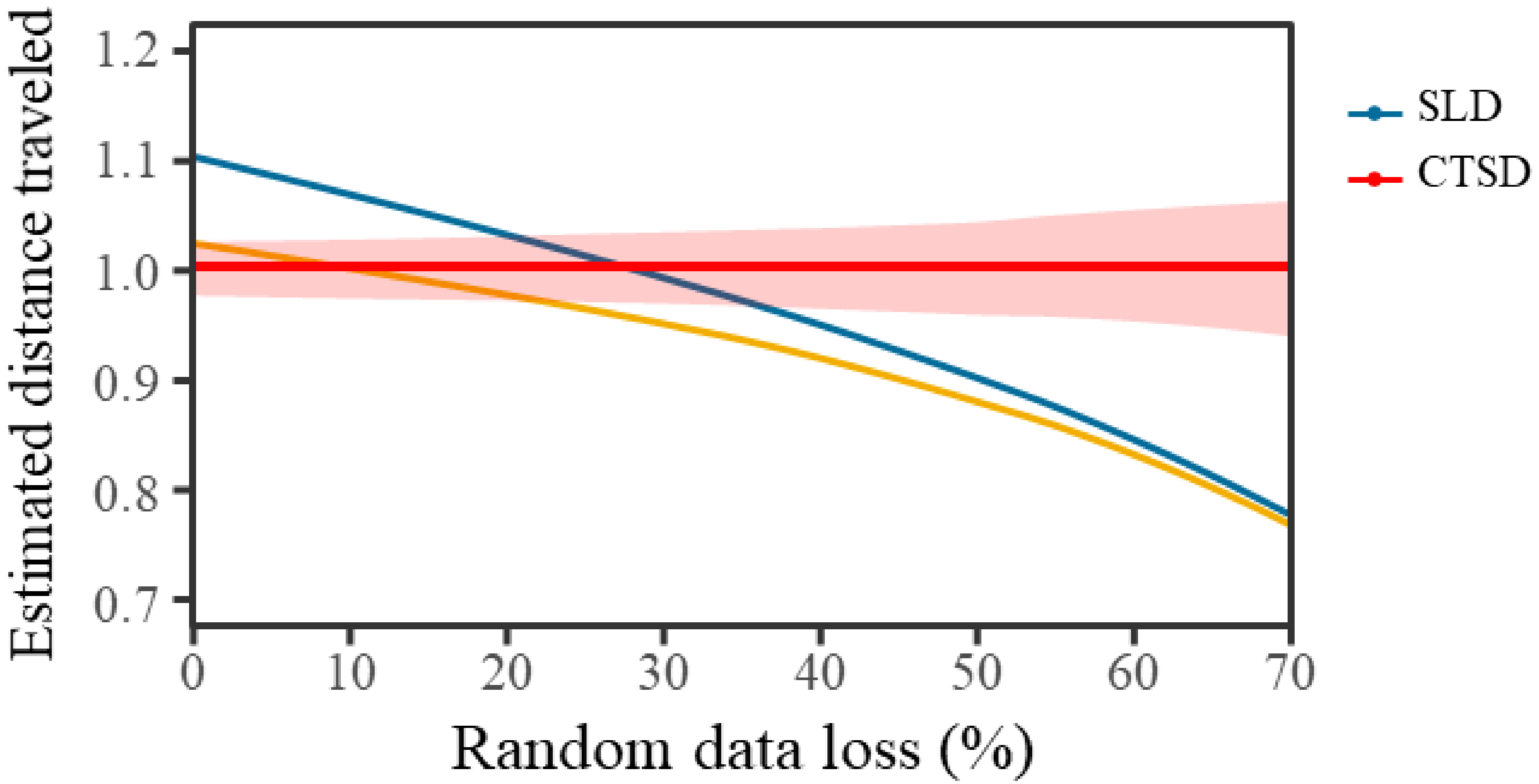
Estimating speed and distance

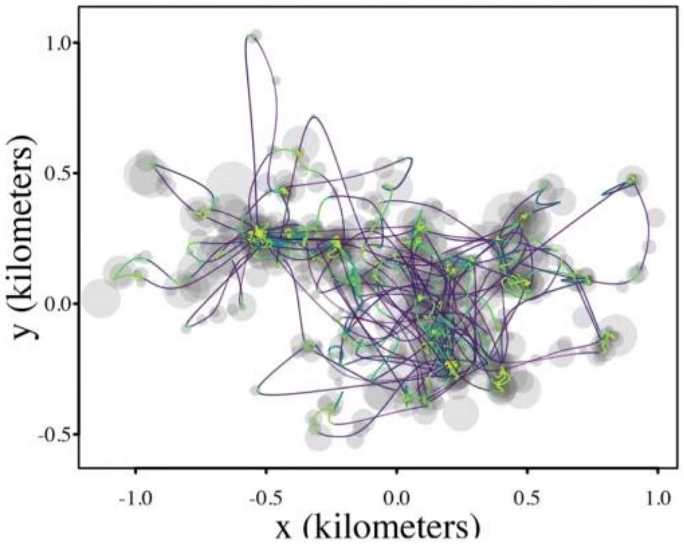


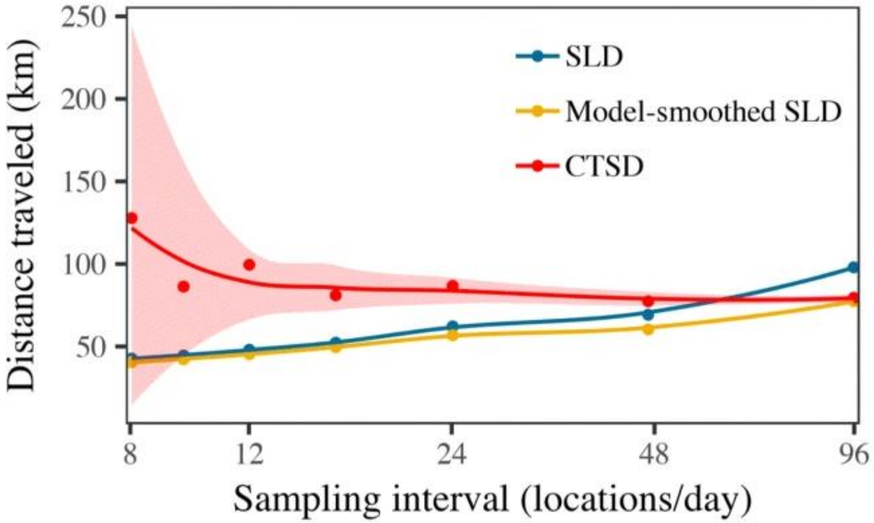
Estimating speed and distance

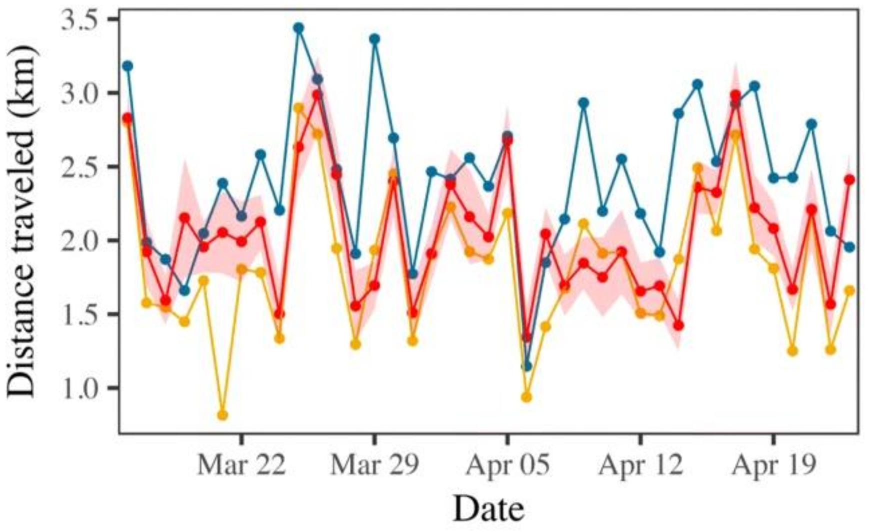


Applications

Sampling *frequency (max)*
1 fix every 15 minutes







Chuck Homler

LC

White-nosed coati
(Nasua narica)

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Applications

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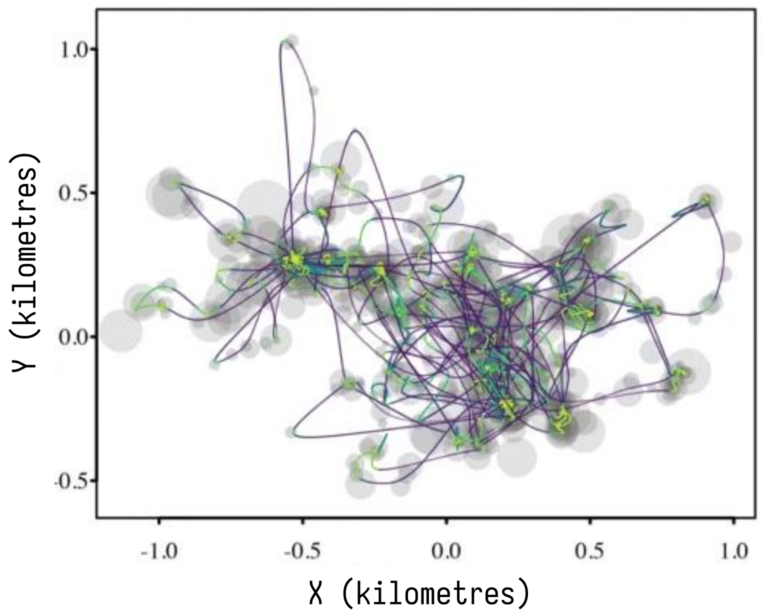

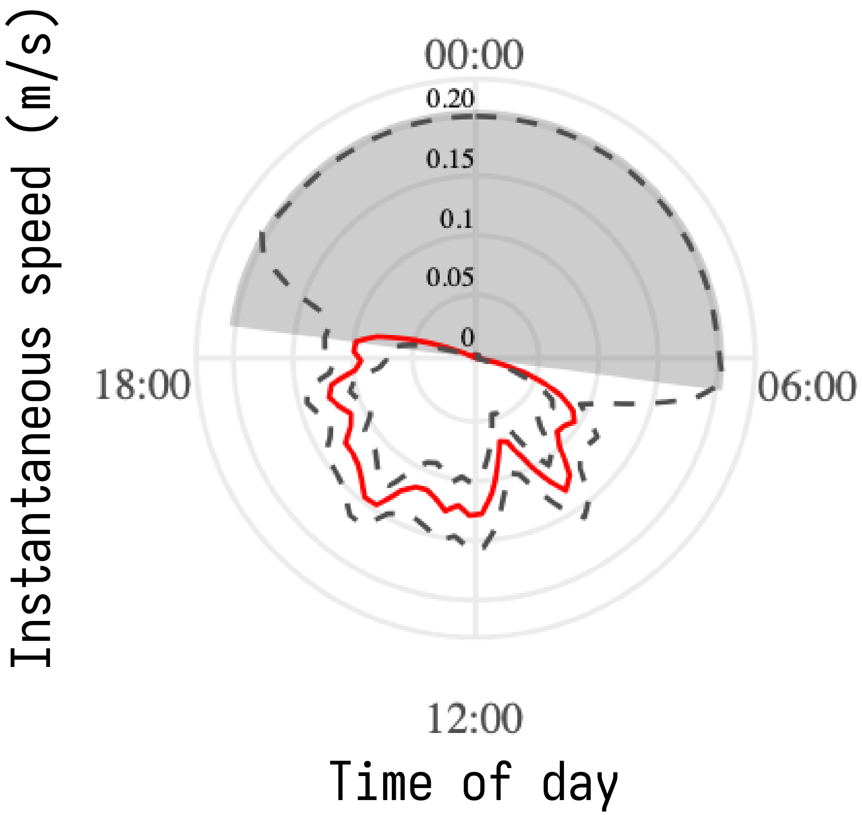

White-nosed coati
(Nasua narica)

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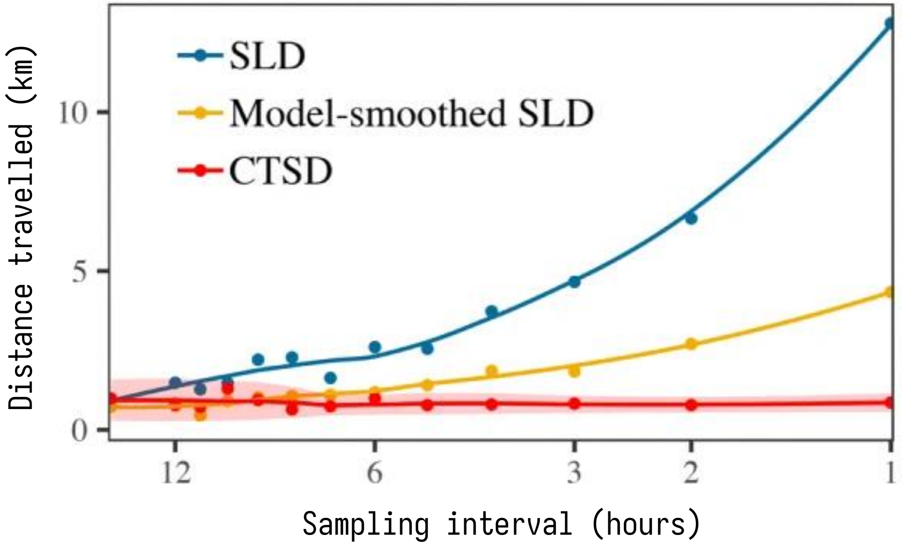
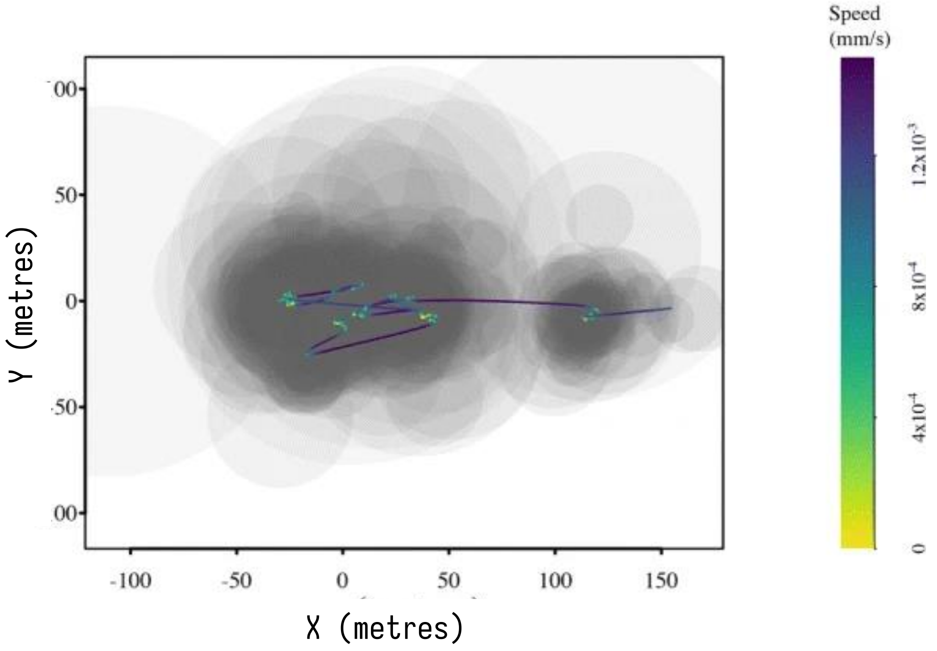
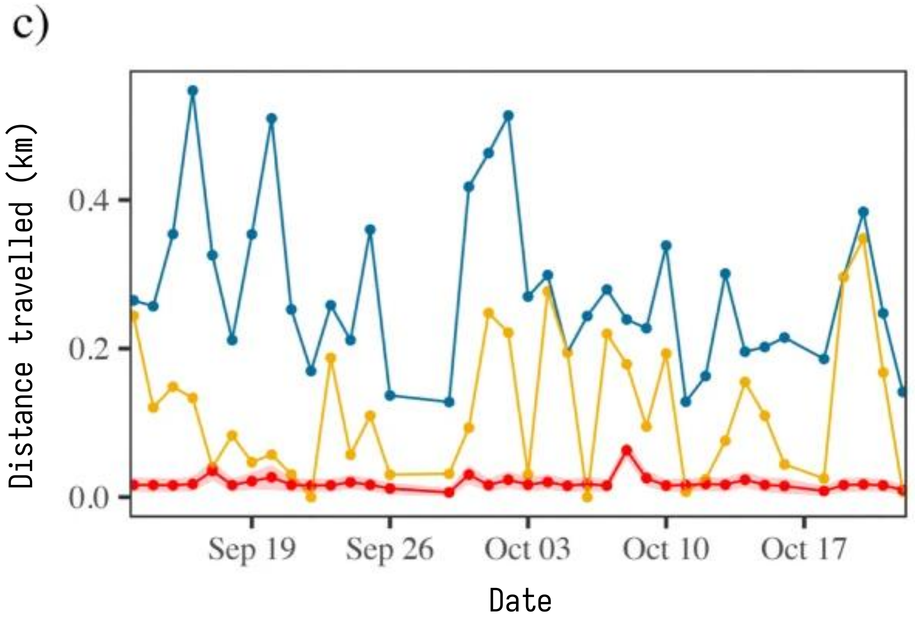
Wood turble

(Glyptemys insculpta)







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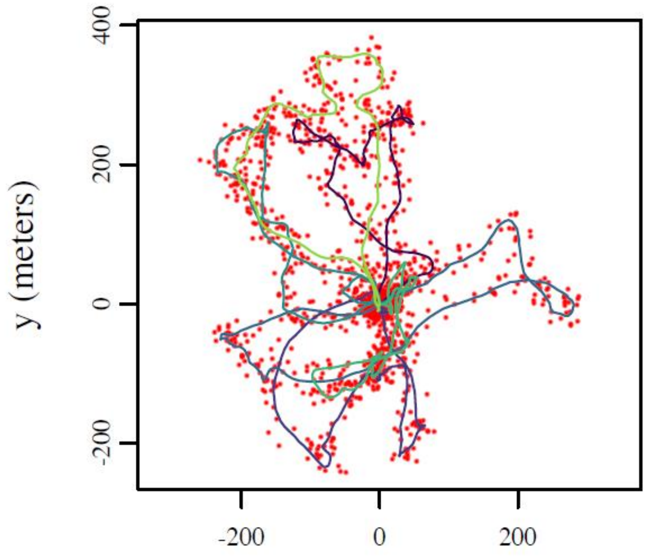
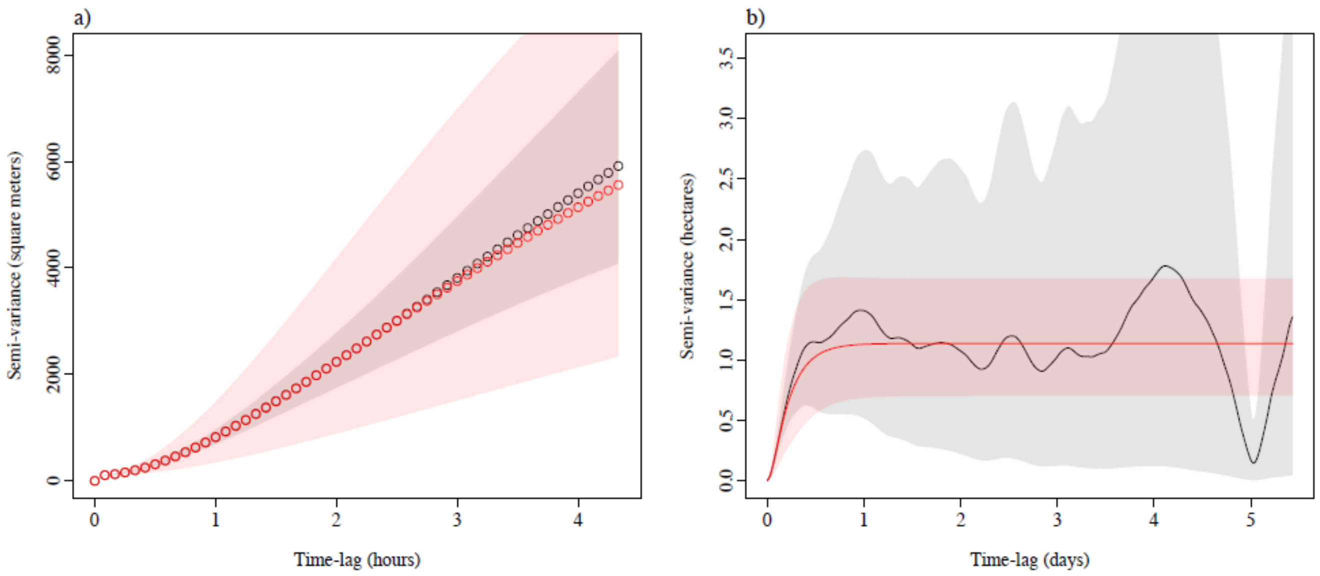
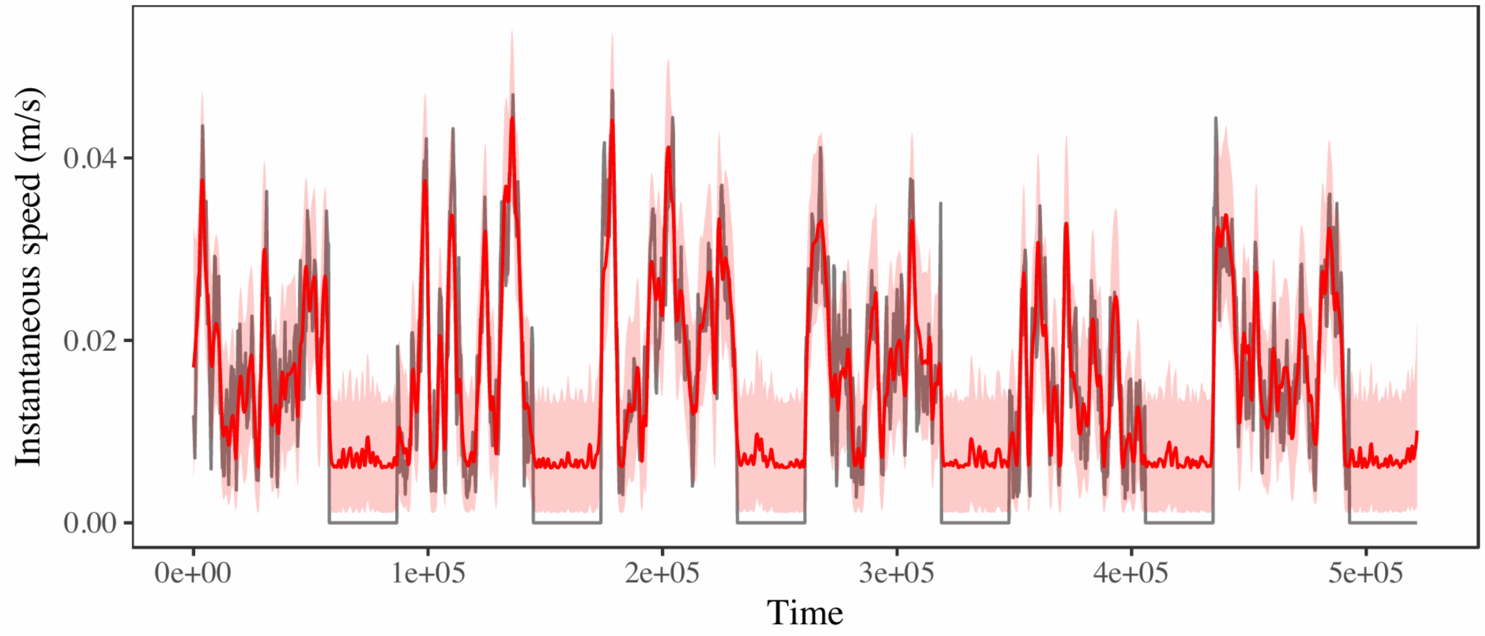
Limitations

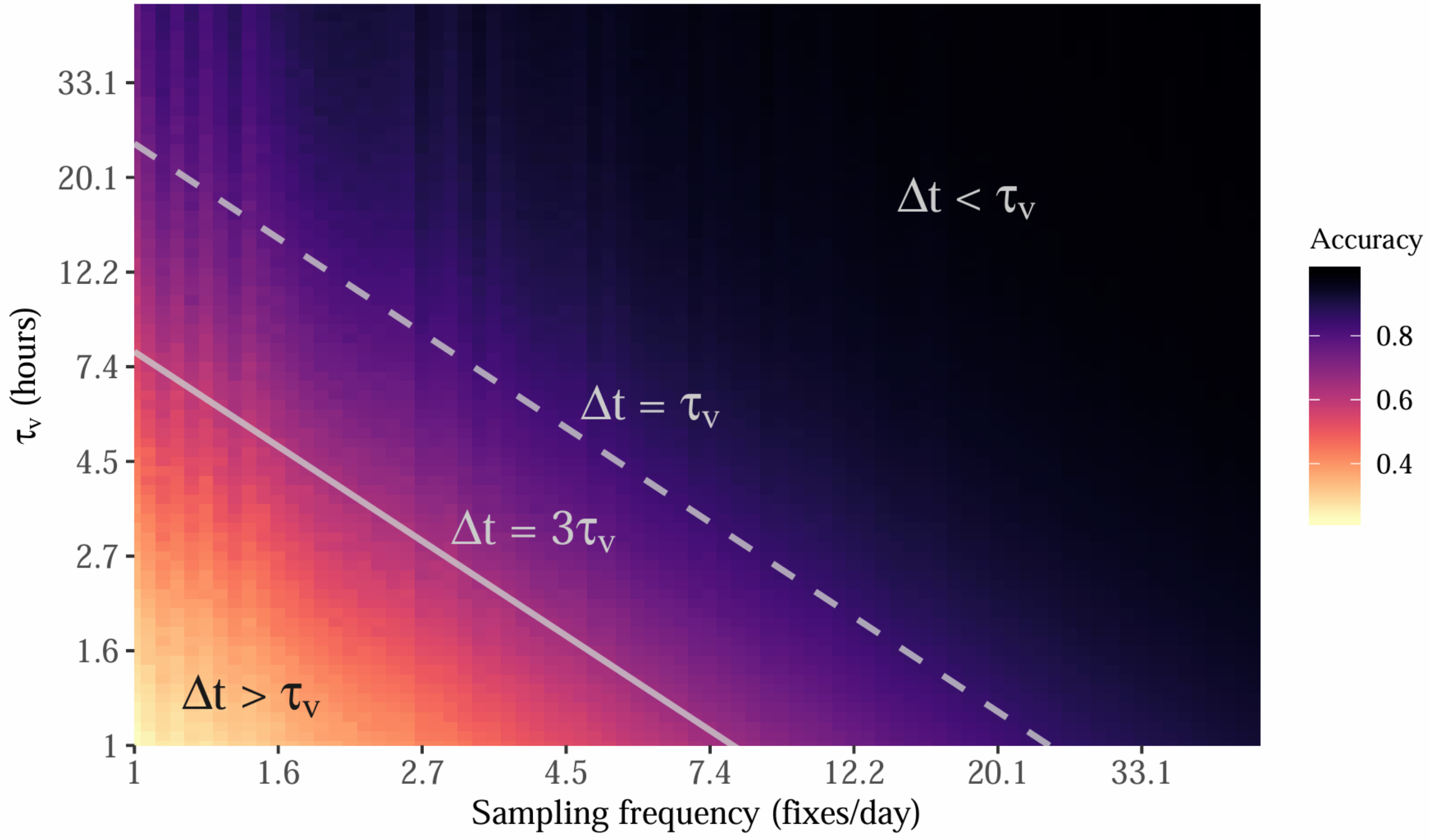






Conditioned on a fitted model.

Requires a correlated velocity model



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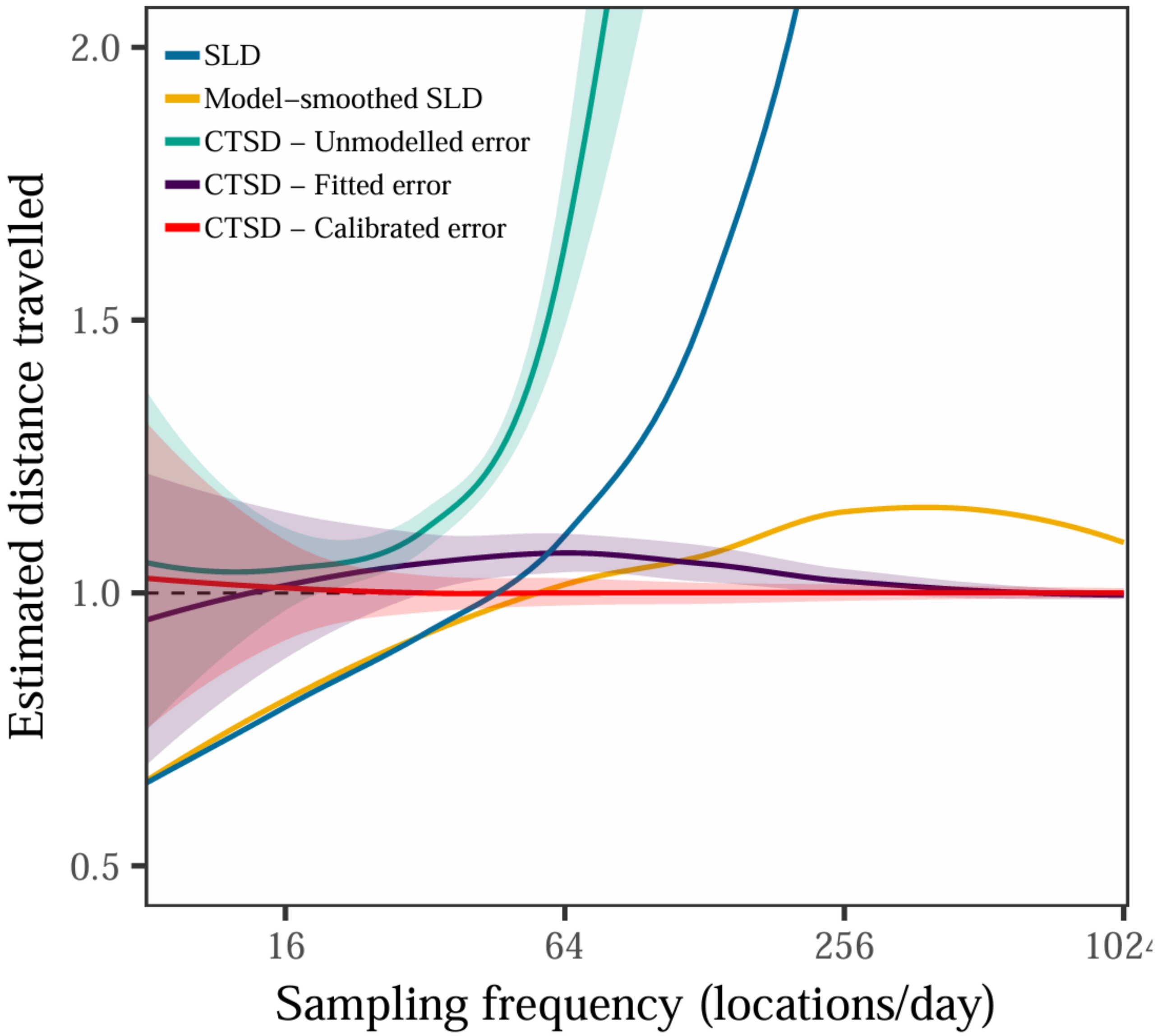
Requires an error model



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ctmm_speed.R